

Packing a Train

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Abstract

A cargo-loading problem with many constraints is considered. General techniques are discussed, and it is argued that a strictly computational approach would not succeed in this case. Specific observations about the details of the approach facilitate narrowing down an exact solution with only 6 mm of wasted length in the two flatcars. This solution is proven to be correct. Different optimal solutions are discussed and selected. They are then analyzed for sensitivity to the given data. Finally, one of the constraints is relaxed to obtain a larger problem, which is solved using symmetries and number-theoretic arguments. General methods to reduce computations in brute-force algorithms are discussed.

1. The Problem to be Solved

We are given 48 different crates of 7 distinct types. All crates have the same width and height, but the different types have different thicknesses and masses. Complete data for all the crates is given in Table 1. We are to load two railroad flatcars of given length from among those crates provided. In so doing, we must respect the mass capacities of the two flatcars, while also heeding a certain special restriction on the total length of crates of types 5, 6, and 7 which may be used (due to a subsequent trucking restriction). Our assignment is to load the cars in such a way as to minimize wasted floor space.

The problem as stated leaves little room for interpretation and requires few assumptions. We have assumed the following:

- 1) The data provided are exact (but see the section on stability below).

Table 1.
Crate data.

Crate type	Number available	Thickness (cm)	Mass (kg)
1	8	48.7	2000
2	7	52.0	3000
3	9	61.3	1000
4	6	72.0	500
5	6	48.7	4000
6	4	52.0	2000
7	8	64.0	1000

2) The distribution of the wasted floor space between the two cars is unimportant, i.e., we are concerned only with the total wasted space.

3) The order in which the crates are packed is unimportant.

4) The two flatcars are identical.

The problem may now be modeled as follows. Deciding how to load the cars amounts to specifying a loading configuration, i.e., nonnegative integers n_i and n'_i , for $i = 1, 2, \dots, 7$, where n_i and n'_i are the numbers of crates of type i to be loaded onto cars 1 and 2, respectively. For a configuration to be *admissible*, it must satisfy certain inequalities. Let h_i be the number of crates of type i available. If l_i and w_i are the thickness and mass of crates of type i , and if $L_{\max}$ ($= 10.2$ m) is the length of each car, $W_{\max}$ ($= 40$ metric tons) is the mass capacity of each car, and $S_{\max}$ ($= 302.7$ cm) is the maximum total length of crates of types 5, 6, and 7 permitted, then the configuration must satisfy:

R1. The length restriction on car 1: $\sum_{i=1}^7 n_i l_i \leq L_{\max}$;

R2. The length restriction on car 2: $\sum_{i=1}^7 n'_i l_i \leq L_{\max}$;

R3. The mass restriction on car 1: $\sum_{i=1}^7 n_i w_i \leq W_{\max}$;

R4. The mass restriction on car 2: $\sum_{i=1}^7 n'_i w_i \leq W_{\max}$;

R5. The "special restriction" on crates of types 5, 6, and 7:

$$\sum_{i=5}^7 (n_i + n'_i) l_i \leq S_{\max}; \text{ and}$$

R6. The availability restrictions: For each i , $1 \leq i \leq 7$, we require

$$n_i + n'_i \leq h_i.$$

The problem may then be stated as having to find the configuration satisfying these constraints which maximizes the total length

$$L_{\text{tot}} = \sum_{i=1}^7 (n_i + n'_i) l_i.$$

2. Introduction

The solution to a problem to maximize a certain quantity subject to certain constraints consists of values of the various parameters such that the constraints are satisfied, and such that the desired quantity is as large as possible. If the parameters in question may vary *continuously* within certain intervals, then the problem may be attacked by the method of Lagrange multipliers (or, for linear systems, by linear-programming methods). However, if the parameter values are *discrete*, other methods are more appropriate. The classical *cargo-loading*, or *knapsack*, problem consists of, for each of n different types of objects, their (non-negative) *value* v_i and (positive) *weight* w_i . The object is to find nonnegative integers $k_1, k_2, \dots, k_n$, which maximize the total value $\sum_{i=1}^n k_i v_i$, subject to the constraint that the total weight $\sum_{i=1}^n k_i w_i$ is not larger than some constant W . This problem has a natural generalization to m constraints, in which case each of m "weights" $w_{i,1}, w_{i,2}, \dots, w_{i,m}$ must be given for each of the n objects, and it is then demanded that their "*j-weight*" $\sum_{i=1}^n k_i w_{i,j}$ be no larger than a constant W_j , for $j = 1, 2, \dots, m$. The m -constraint problem may be approached by at least two general methods, each of which will in theory always produce an optimal solution.

One of these is the so-called *brute force* method. This method makes use of the fact that discrete, bounded parameters may take on only a finite number of different values. Thus, it is in theory possible to consider every conceivable configuration in turn and check if it satisfies the given restrictions. If it does, it can be compared to the previous best configuration, and the better of the two stored. One need only remember the optimal configuration found to date; and when the procedure terminates, the overall optimal configuration will be in hand. This solution, while not elegant, is guaranteed to succeed, and requires very little memory space when implemented on a computer. However, for large problems it takes a lot of time to execute, and in some cases it may be prohibitive.

The second general method involves the concept of *dynamic programming*. In this approach, a somewhat more general problem is solved for a more restricted class of objects. Define $f_j(y_1, y_2, \dots, y_m)$ to be the largest possible total value obtained by using only objects $1, 2, \dots, j$ and such that each k -weight is $\leq y_k$, defined for all possible y_k such that $0 \leq y_k \leq W_k$. Then f_1 is given by:

$$f_1(y_1, y_2, \dots, y_m) = \max_{\substack{n_1 \\ 0 \leq n_1 w_{1,k} \leq y_k \\ k=1, 2, \dots, m}} n_1 v_1.$$

The function f_j may then be computed recursively from f_{j-1} by:

$$f_j(y_1, y_2, \dots, y_m) = \max_{\substack{n_j \\ 0 \leq n_j w_{j,k} \leq y_k \\ k=1, 2, \dots, m}} \left[n_j v_j + f_{j-1}(y_1 - n_j w_{j,1}, y_2 - n_j w_{j,2}, \dots, y_m - n_j w_{j,m}) \right],$$

The answer to the problem is then given by

$$f_n(W_1, W_2, \dots, W_m).$$

If the functions $f_1, f_2, \dots, f_n$ are computed and stored in succession, the overall time to execute this algorithm will generally be less than that required for brute force. However, in the event that there are many possible values of the y_j to be considered, the amount of memory required to store each f_j will be substantial, and the time required to compute each f_j completely will correspondingly increase.

3. Analysis of the Problem

The problem we are given falls into the general class of an m -constraint cargo-loading scenario. Thus, it was reasonable to consider the feasibility of either a *brute force* or *dynamic programming* approach as discussed in the introduction.

An order-of-magnitude estimate of the amount of time required to use brute force is obtained as follows. Each of the parameters n_i and n'_i may vary such that $n_i + n'_i \leq h_i$. The number of possible ordered pairs (n_i, n'_i) is thus given by

$$\sum_{k=0}^{h_i} (k+1) = \sum_{k=1}^{h_i+1} k = \frac{1}{2}(h_i+1)(h_i+2).$$

Hence, the number of configurations which need to be considered is

$$\prod_{i=1}^7 \frac{1}{2}(h_i+1)(h_i+2),$$

which is about 4.7×10^{10} . This is far too large a number to allow direct computation by this method. Without somehow cutting down the amount of work required, the brute-force method cannot be used.

"...brute-force cannot be used...."

The dynamic programming method presented difficulties because of the large number of constraints involved in the problem. Recalling that restriction R6 above amounts to 7 constraints, there is a total of 12 constraints involved in the problem. Dynamic programming would thus require that the f_i be functions of 12 variables. However, by slightly modifying the general dynamic-programming algorithm, we can accommodate the R6 constraints directly, and thus need not let the f_j depend on them. More precisely, let $f_j(L, L', W, W', S)$ be the largest total length (i.e., value) of a packing that uses only crates of types $1, 2, \dots, j$, and for which the total length of crates in car 1 is $\leq L$, the total length of crates in car 2 is $\leq L'$, the total mass of crates in car 1 is $\leq W$, the total mass of crates in car 2 is $\leq W'$, and (when $j = 5, 6$, or 7) the total length of crates of types $5, 6$, and 7 in both cars is $\leq S$. We may then compute f_1 by:

$$f_1 = \max_{\substack{n_1, n'_1 \\ n_1 + n'_1 \leq h_1 \\ n_1 l_1 \leq L \\ n'_1 l_1 \leq L' \\ n_1 w_1 \leq W \\ n'_1 w_1 \leq W'}} (n_1 + n'_1) l_1,$$

The function f_j can then be computed from f_{j-1} by

$$f_j = \max_{\substack{n_j, n'_j \\ n_j + n'_j \leq h_j \\ n_j l_j \leq L \\ n'_j l_j \leq L' \\ n_j w_j \leq W \\ n'_j w_j \leq W' \\ (n_j + n'_j) l_j \leq S \text{ if } j=5, 6, 7}} [(n_j + n'_j) l_j + f_{j-1}(L - n_j l_j, L' - n'_j l_j, W - n_j w_j, W' - n'_j w_j, S - A_j)],$$

where $A_j = n_j l_j$ if $j = 5, 6$, or 7 , and is zero otherwise.

The overall optimum length is then given by

$$f_7(L_{\max}, L_{\max}, W_{\max}, W_{\max}, S_{\max}).$$

This implementation allows the restrictions R6 above to be handled immediately and thus decreases the amount of memory required to store the values of the functions f_j . Despite this improvement, the resulting functions f_j must still be computed for an extremely large number of input vectors. In fact, to accommodate all of the data, lengths must be measured in millimeters, which sets $L_{\max}$ to 10200 and $S_{\max}$ to 3027. We can measure mass in units of 500 kg, in which case $W_{\max}$ is 80. This means that the array required to store the values of a typical f_j has dimensions $10201 \times 10201 \times 81 \times 81 \times$

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"...The approach we take is to reduce the size of the problem."

3028, and thus requires about 2.0×10^{15} computations, which is prohibitively large. Hence, without dramatic improvements being made on the basic algorithm, dynamic programming would appear to rule itself out.

The approach we take is to reduce the size of the problem so that one of the above methods (or perhaps some other) will be usable. We do this by reducing the number of configurations that are worth considering.

Our problem has a natural symmetry: If each n_i is exchanged with n'_i , the constraints and L_{tot} are unchanged. This corresponds to relabelling the two flatcars. Thus, when searching for the optimal solution, we need only consider one of each of these pairs of configurations. One simple way to almost do this is to specify that $n_2 \leq n'_2$; this cuts down the number of configurations that need checking by almost a factor of 2.

The second, and *decisive*, reduction for our problem comes from considering the "strengths" of the given constraints. If a constraint is hard to satisfy (i.e., *strong*), then it may have certain necessary (but perhaps not sufficient) implications for its satisfaction. If a constraint is easy to satisfy (i.e., *weak*), there may be a simple conditions under which the constraint will always be satisfied.

As a simple application, consider the restrictions **R5** and **R6** above. **R5** necessarily implies that

$$(n_7 + n'_7)l_7 \leq \sum_{i=5}^7 (n_i + n'_i)l_i \leq S_{\max},$$

since all these variables are nonnegative. Substituting the values of l_7 and $S_{\max}$ yields the inequality $n_7 + n'_7 \leq 4$, i.e., at most 4 crates of type 7 can be used. This replaces the weaker availability constraint **R6** for $i = 7$, which states that $n_7 + n'_7 \leq 8$.

Consider for the moment a configuration for which **R5** and **R6** are satisfied. Since $l_5, l_6, l_7 \geq 48.7$ cm, **R5** implies that $\sum_{i=5}^7 (n_i + n'_i) 48.7 \text{ cm} \leq \sum_{i=5}^7 (n_i + n'_i)l_i \leq S_{\max} = 302.7$ cm. Thus $\sum_{i=5}^7 (n_i + n'_i) \leq 6$, i.e., the configuration uses a total of at most 6 crates of types 5, 6, and 7. This has important implications for **R1** to **R4**. Since $w_5, w_6, w_7 \leq 4000$ kg, the total mass contribution from crates of types 5, 6, and 7 is $\sum_{i=5}^7 (n_i + n'_i)w_i \leq 6 \times 4000 = 24000$ kg. Thus the configuration has a total mass of $\sum_{i=1}^7 (n_i + n'_i)w_i \leq \sum_{i=1}^4 h_i w_i + \sum_{i=5}^7 (n_i + n'_i)w_i \leq 49000 + 24000 = 73000$ kg. Since the total mass capacity of the two cars is 80000 kg, it is impossible to violate the mass capacity of both cars. In fact, one car must have at least 3500 kg of unused capacity. Also, the total length used for the configuration is $\sum_{i=1}^7 (n_i + n'_i)l_i \leq \sum_{i=1}^4 h_i l_i + \sum_{i=5}^7 (n_i + n'_i)l_i \leq 1737.3 +$

302.7 (by R5) = 2040.0 cm. Since 2040 cm is the total available space on both cars, it is impossible to overfill them both, i.e., only one of R1 and R2 can be violated.

4. The Solution

As a result of the observation that restrictions R5 and R6 are "strong" and restrictions R1 to R4 "weak" (see end of previous section), it is believed that the optimal configuration will use a lot of crates of types 1, 2, 3, and 4. A little bit of calculation makes this notion precise in the form of the following remarkable theorem.

Theorem 1. Any optimal (admissible) configuration in the given problem uses *all* of the available crates of types 1, 2, 3, and 4.

We prove the theorem with the aid of the following lemma.

Lemma. Any admissible configuration *not* using all available crates of types 1, 2, 3, and 4 has $L_{\text{tot}} \leq (2040 - 48.7) = 1991.3$ cm.

Proof of Lemma: Consider any configuration of the specified type. Then there is some i , $i = 1, 2, 3$, or 4, such that $n_i + n'_i \leq h_i - 1$. Thus,

$$\begin{aligned} L_{\text{tot}} &= \sum_{j=1}^7 (n_j + n'_j) l_j \\ &\leq \sum_{j=1}^4 (n_j + n'_j) l_j + 302.7 \quad (\text{by constraint R5}) \\ &\leq \sum_{j=1}^4 (h_j - \delta_{ij}) l_j + 302.7 \quad (\text{by assumption}) \\ &= \sum_{j=1}^4 h_j l_j - l_i + 302.7 \\ &= 1737.3 - l_i + 302.7 \quad (\text{by calculation}) \\ &= 2040.0 - l_i \\ &\leq 2040.0 + \min_{k=1}^4 l_k \\ &= 2040.0 - 48.7 \\ &= 1991.3 \text{ cm. } \square \end{aligned}$$

Proof of Theorem 1: By the lemma, we need only show that the largest possible value of L_{tot} is more than 1991.3 cm. To do this, it suffices to find a single admissible configuration that satisfies $L_{\text{tot}} >$

1991.3 cm. This can be done easily by inspection. The first such configuration that we found (by hand) was:

$$\begin{aligned} n_1 &= 4 & n_2 &= 2 & n_3 &= 6 & n_4 &= 3 & n_5 &= n_6 &= 0 & n_7 &= 2 \\ n'_1 &= 4 & n'_2 &= 5 & n'_3 &= 3 & n'_4 &= 3 & n'_5 &= n'_6 &= 0 & n'_7 &= 2. \end{aligned}$$

A straightforward calculation shows that for this configuration, the masses in cars 1 and 2 are 23.5 and 29.5 metric tons, respectively. The lengths taken up in the cars are 1010.6 and 982.7 cm, respectively. The total length of crates of type 5, 6, and 7 is 256.0 cm. Thus, all constraints are satisfied. The total length of $L_{tot} = 1993.3$ cm. $\square$

Because of **Theorem 1**, we need only check those configurations that use all crates of types 1, 2, 3, and 4. This greatly reduces the amount of time necessary to implement a brute-force type of approach. All that is required is to compute all combinations of crates of types 5, 6, and 7 which satisfy constraint **R5**. From among these combinations, we need only find one with maximal total length which still allows all crates of types 1, 2, 3, and 4 to be used. In fact, the proof of **Theorem 1** shows that the optimal solution is larger than 1991.3 cm, so we need only consider those combinations for which $\sum_{i=1}^4 h_i l_i + \sum_{i=5}^7 (n_i + n'_i) l_i > 1991.3$. Also, knowing that all crates of types 1 to 4 must be used means we can assume $n_i + n'_i = h_i$, for $i = 1, 2, 3, 4$, which cuts down substantially on the amount of checking that must be done.

This searching was done with the aid of a Sun Microsystems computer, in two stages. First, all combinations of crates of types 5, 6, and 7 which satisfy constraint **R5** and which are within $2040.0 - 1991.3 = 48.7$ cm of S_{max} were found. These combinations are displayed in Table 2. As can be seen from the table, the largest total length of any such combination is 302.1 cm. This result immediately implies the following theorem.

Theorem 2. No admissible configuration has a total length of more than 2039.4 cm.

Proof: Consider any admissible configuration. Then $L_{tot} = \sum_{i=1}^4 (n_i + n'_i) l_i$. But from the above discussion, $\sum_{i=5}^7 (n_i + n'_i) l_i \leq 302.1$ cm. Also, $\sum_{i=1}^4 (n_i + n'_i) l_i \leq \sum_{i=1}^4 h_i l_i = 1737.3$ cm (by previous calculation). Hence, $\sum_{i=1}^7 (n_i + n'_i) l_i \leq 1737.3 \text{ cm} + 302.1 \text{ cm} = 2039.4$ cm. $\square$

The combinations found were then sorted in decreasing L_{tot} using a UNIX utility. The sorted combinations were fed into a second program, which tried to find, for each combination, a configuration including it and all crates of types 1 to 4, such that all constraints were satisfied. The results were dramatic. Even for the largest combination considered, namely using 3 crates of each of

Table 2.
All configurations $(C_5C_6C_7)$ with $3027 - 487 < L_{\text{tot}} \leq 3027$ mm.

L_{tot}	C_5	C_6	C_7
2560	0	0	4
2567	0	0	4
2588	4	0	1
2621	3	1	1
2654	2	2	1
2687	1	3	1
2720	0	4	1
2741	3	0	2
2774	2	1	2
2807	1	2	2
2840	0	3	2
2894	2	0	3
2922	6	0	0
2927	1	1	3
2955	5	1	0
2960	0	2	3
2988	4	2	0
3021	3	3	0

types 5 and 6 and none of type 7, it was found that there were many different configurations that satisfied all of the given constraints. This result, together with Theorem 2, solves the problem. We present the solution in the form of a theorem.

Theorem 3. The largest length that can be used in cars 1 and 2 while satisfying the given constraints is 20.394 m, or 2039.4 cm (so that 6 mm is left unused). One configuration (of many) yielding this length is:

$$\begin{aligned} n_1 = 0, \quad n_2 = 7, \quad n_3 = 9, \quad n_4 = 0, \quad n_5 = 0, \quad n_6 = 2, \quad n_7 = 0, \\ n'_1 = 8, \quad n'_2 = 0, \quad n'_3 = 0, \quad n'_4 = 6, \quad n'_5 = 3, \quad n'_6 = 1, \quad n'_7 = 0. \end{aligned}$$

Proof: The given configuration has length 10.197 m in each car, with total length from crates of type 5, 6, and 7 equal to 302.1 cm. The masses of crates in cars 1 and 2 are 34.0 and 33.0 metric tonnes, respectively. Thus, all restrictions are satisfied, and the result follows from Theorem 2. $\square$

All optimal solutions (those with maximal total length) are presented in Table 3. Each of these configurations uses the same number of crates of each type (though not necessarily the same number in each car), and thus each has the same total length. This length is 2039.4 cm, or 20.394 m. The total mass is also fixed, at 67000 kg or 67 tonnes.

"... 6 mm
is left
unused...."

Table 3
All optimal configurations ($L_0 = 2039.4$ cm).

Table 3 All optimal configurations ($L_n = 2039.4$ cm).													
#	Configuration (C1 C2 C3 C4 C5 C6 C7)							Length Occupied			Weight Occupied		
	Car 1							Car 1	Car 2	Total	Car 1	Car 2	Total
1	6	2	9	1	0	0	0	10199	10195	20394	27500	39500	67000
2	7	3	5	3	0	0	0	10194	10200	20394	29500	37500	67000
3	4	7	4	3	0	0	0	10200	10194	20394	34500	32500	67000
4	0	7	6	4	0	0	0	10198	10196	20394	29000	38000	67000
5	7	2	5	3	0	1	0	10194	10200	20394	28500	38500	67000
6	4	6	4	3	0	1	0	10200	10194	20394	33500	33500	67000
7	0	6	6	4	0	1	0	10198	10196	20394	28000	39000	67000
8	5	7	0	5	0	1	0	10195	10199	20394	35500	31500	67000
9	0	7	9	0	0	2	0	10197	10197	20394	34000	33000	67000
10	7	1	5	3	0	2	0	10194	10200	20394	27500	39500	67000
11	4	5	4	3	0	2	0	10200	10194	20394	32500	34500	67000
12	0	5	6	4	0	2	0	10198	10196	20394	27000	40000	67000
13	5	6	0	5	0	2	0	10195	10199	20394	34500	32500	67000
14	0	6	9	0	0	3	0	10197	10197	20394	33000	34000	67000
15	4	4	4	3	0	3	0	10200	10194	20394	31500	35500	67000
16	5	5	0	5	0	3	0	10195	10199	20394	33500	33500	67000
17	5	2	9	1	1	0	0	10199	10195	20394	29500	37500	67000
18	6	3	5	3	1	0	0	10194	10200	20394	31500	35500	67000
19	3	7	4	3	1	0	0	10200	10194	20394	36500	30500	67000

20	5	1	9	1	1	1	0	3	6	0	5	2	2	0	10199	10195	20394	28500	38500	67000
21	6	2	5	3	1	1	0	2	5	4	3	2	2	0	10194	10200	20394	30500	36500	67000
22	3	6	4	3	1	1	0	5	1	5	3	2	2	0	10200	10194	20394	35500	31500	67000
23	4	7	0	5	1	1	0	4	0	9	1	2	2	0	10195	10199	20394	37500	29500	67000
24	5	0	9	1	1	2	0	3	7	0	5	2	1	0	10199	10195	20394	27500	39500	67000
25	6	1	5	3	1	2	0	2	6	4	3	2	1	0	10194	10200	20394	29500	37500	67000
26	3	5	4	3	1	2	0	5	2	5	3	2	1	0	10200	10194	20394	34500	32500	67000
27	4	6	0	5	1	2	0	4	1	9	1	2	1	0	10195	10199	20394	36500	30500	67000
28	6	0	5	3	1	3	0	2	7	4	3	2	0	0	10194	10200	20394	28500	38500	67000
29	3	4	4	3	1	3	0	5	3	5	3	2	0	0	10200	10194	20394	33500	33500	67000
30	4	5	0	5	1	3	0	4	2	9	1	2	0	0	10195	10199	20394	35500	31500	67000
31	4	2	9	1	2	0	0	4	5	0	5	1	3	0	10199	10195	20394	31500	35500	67000
32	5	3	5	3	2	0	0	3	4	4	3	1	3	0	10194	10200	20394	33500	33500	67000
33	2	7	4	3	2	0	0	6	0	5	3	1	3	0	10200	10194	20394	38500	28500	67000
34	4	1	9	1	2	1	0	4	6	0	5	1	2	0	10199	10195	20394	30500	36500	67000
35	5	2	5	3	2	1	0	3	5	4	3	1	2	0	10194	10200	20394	32500	34500	67000
36	2	6	4	3	2	1	0	6	1	5	3	1	2	0	10200	10194	20394	37500	29500	67000
37	3	7	0	5	2	1	0	5	0	9	1	1	2	0	10195	10199	20394	39500	27500	67000
38	4	0	9	1	2	2	0	4	7	0	5	1	1	0	10199	10195	20394	29500	37500	67000
39	5	1	5	3	2	2	0	3	6	4	3	1	1	0	10194	10200	20394	31500	35500	67000
40	2	5	4	3	2	2	0	6	2	5	3	1	1	0	10200	10194	20394	36500	30500	67000
41	3	6	0	5	2	2	0	5	1	9	1	1	1	0	10195	10199	20394	38500	28500	67000
42	5	0	5	3	2	3	0	3	7	4	3	1	0	0	10194	10200	20394	30500	36500	67000
43	2	4	4	3	2	3	0	6	3	5	3	1	0	0	10200	10194	20394	35500	31500	67000
44	3	5	0	5	2	3	0	5	2	9	1	1	0	0	10195	10199	20394	37500	29500	67000

Table 3. (continued)

#	Configuration (C1 C2 C3 C4 C5 C6 C7)							Table 3. (continued)			Weight Occupied									
	Car 1							Length Occupied			Car 1	Car 2	Total							
	Car 2							Car 1	Car 2	Total										
45	3	2	9	1	3	0	0				5	5	0	5	0	3	0	10199	10195	20394
46	4	3	5	3	3	0	0	4	4	4	3	0	3	0	10194	10200	20394	35500	31500	67000
47	8	1	0	6	3	0	0	0	6	9	0	0	3	0	10197	10197	20394	34000	33000	67000
48	3	1	9	1	3	1	0	5	6	0	5	0	2	0	10199	10195	20394	32500	34500	67000
49	8	2	3	2	3	1	0	0	5	6	4	0	2	0	10196	10198	20394	40000	27000	67000
50	4	2	5	3	3	1	0	4	5	4	3	0	2	0	10194	10200	20394	34500	32500	67000
51	1	6	4	3	3	1	0	7	1	5	3	0	2	0	10200	10194	20394	39500	27500	67000
52	8	0	0	6	3	1	0	0	7	9	0	0	2	0	10197	10197	20394	33000	34000	67000
53	3	0	9	1	3	2	0	5	7	0	5	0	1	0	10199	10195	20394	31500	35500	67000
54	8	1	3	2	3	2	0	0	6	6	4	0	1	0	10196	10198	20394	39000	28000	67000
55	4	1	5	3	3	2	0	4	6	4	3	0	1	0	10194	10200	20394	33500	33500	67000
56	1	5	4	3	3	2	0	7	2	5	3	0	1	0	10200	10194	20394	38500	28500	67000
57	8	0	3	2	3	3	0	0	7	6	4	0	0	0	10196	10198	20394	38000	29000	67000
58	4	0	5	3	3	3	0	4	7	4	3	0	0	0	10194	10200	20394	32500	34500	67000
59	1	4	4	3	3	3	0	7	3	5	3	0	0	0	10200	10194	20394	37500	29500	67000
60	2	5	0	5	3	3	0	6	2	9	1	0	0	0	10195	10199	20394	39500	27500	67000

Table 4
Balanced solutions

#	Configurations C1 C2 C3 C4 C5 C6 C7							L_1	L_2	$L_{[6]}$	L_{tot}	W_1	W_2	$W_{[6]}$	W_{tot}	
	Car 2															
1	0	6	9	0	0	3	0	0	10197	10197	0	20394	33000	34000	1000	67000
2	0	5	4	5	1	2	0	0	10179	19182	3	20361	29500	39500	10000	69000
3	0	1	7	6	0	2	0	3	10171	10162	9	20333	17000	39000	22000	56000
4	0	7	9	0	1	1	0	0	10164	10164	0	20328	36000	35000	1000	71000
5	0	4	8	3	1	1	0	3	10151	10149	2	20300	27500	30500	3000	58000
6	0	3	3	6	5	0	0	0	10154	10141	13	20295	35000	38000	3000	73000
7	0	7	0	5	2	0	3	0	10134	10133	1	20267	34500	25500	9000	60000
8	0	1	5	6	0	3	1	1	10105	10108	3	20213	18000	39000	21000	57000
9	0	5	7	3	0	2	0	2	10091	10089	2	20180	27500	31500	4000	59000
10	1	5	1	5	2	1	2	0	10074	10073	1	20147	32500	28500	4000	61000
11	0	6	7	3	1	0	0	2	10058	10056	2	20114	30500	32500	2000	63000
12	1	2	6	6	0	1	0	1	10045	10048	3	20093	19000	39000	20000	58000
13	0	2	8	5	1	0	0	3	10031	10029	2	20060	20500	39500	19000	60000
14	0	1	6	6	2	1	0	1	10012	10015	3	20027	22000	40000	18000	62000
15	0	5	9	1	0	1	1	0	9997	9997	0	19994	27500	36500	9000	64000
16	1	4	2	5	4	0	1	0	9981	9980	1	19961	35500	30500	5000	66000
17	0	4	7	5	0	0	0	0	9971	9969	2	19940	21500	39500	18000	61000
18	1	5	0	6	0	0	4	0	9967	9966	1	19933	24000	29000	5000	53000

In addition, for each combination in Table 2, Table 4 provides one of many configurations that allow the combination to "fit" along with all crates of types 1 to 4. In fact, the configurations in Table 4 were chosen on the basis of their sharing the total length most equally between the two flatcars. Section 6, on stability, provides an explanation of why this is desirable.

It should be emphasized that Theorem 3 is an exact solution of the problem. Despite the fact that computers were used to speed up certain calculations, they used only exact integer arithmetic, which involved no approximations or heuristics. Their calculations are a small but solid part of what comprises a *complete* solution including rigorous *proof* of its correctness.

5. Choosing from the Many Optimal Solutions

Because of the many (60) optimal solutions, it seems natural to consider various criteria for selecting the solution to implement. The previous analysis showed that each optimal solution is some arrangement of all the crates of types 1 to 4, three crates each of types 5 and 6, and no crates of type 7. This similarity of the solutions reduces the number of criteria that distinguish among them.

One criterion that is useful is to obtain the largest possible margin of error in satisfying restrictions R1 to R4 above, to accommodate possible measuring errors in crate thicknesses and masses. Since the total length and mass are fixed, this means that it is desirable to spread the excess length and mass capacity as equally as possible between the two cars. Table 3 clearly indicates that there are many optimal solutions that have a comfortable excess mass capacity in both cars. This indicates that it is more important to spread equally the excess *length* than to worry about mass capacities. Choosing the preferred solution on this basis yields (equally well) solutions 9, 14, 47, and 52 in Table 1. Each of these solutions has total length in each car equal to 1019.7 cm, and is thus perfectly balanced with respect to length. As an added bonus, the masses of the two cars are 34000 and 33000 kg, and are thus extremely well balanced also. Thus, we would recommend any of the four solutions 9, 14, 47, and 52 as being superior to any of the others.

6. Stability of the Solution

Although we assumed that the given data are exact, we did consider the sensitivity of our (exact) solution to small changes in

various values. In particular, we considered the effects of slightly varying the restrictive values $L_{\max}$, $W_{\max}$, and $S_{\max}$.

It was first noted that there were many optimal configurations that were quite comfortably within the mass limits sets by $W_{\max}$. This makes it clear that small changes in $W_{\max}$ are of little significance, and that the optimal solutions are not sensitive to its value. Thus, $W_{\max}$ shall not be considered further.

A key component of our exact solution is **Theorem 1**. Note that this theorem continues to apply so long as there is an admissible configuration that satisfies $L_{\text{tot}} \geq 2L_{\max} - 48.7$ cm. Since, for the given values of $L_{\max}$ and $S_{\max}$, there is a configuration with $L_{\text{tot}} = 2L_{\max} - 0.6$ cm, and since the density of obtainable values of L_{tot} near the optimal solution is one every few centimeters, it is apparent that **Theorem 1** will continue to hold as long as neither $L_{\max}$ nor $S_{\max}$ is changed by more than about $(48.7 - 0.6)/2 = 24.05$ cm. We shall assume here that this is the case.

The following observation is important. If $Y = \sum_{i=1}^4 h_i l_i$, then assuming **Theorem 1** holds, the value $\sum_{i=5}^7 (n_i + n'_i)$ is bounded by $M = \min(S_{\max}, L_{\max} - Y)$. The quantity M is thus the only important parameter that depends on $L_{\max}$ and $S_{\max}$ (given the assumption just stated). Hence, there is no loss of information in assuming for simplicity that $S_{\max} = L_{\max} - Y$, and we shall do so in what follows.

Under the above assumptions, if $L_{\max}$ increases (and hence $S_{\max}$ increases similarly), then one would expect the optimum value L_{opt} to go up. This is indeed the case, but the L_{opt} increase is *discrete*. As $L_{\max}$ increases, L_{opt} will at first remain *constant*. When $L_{\max}$ reaches a certain threshold value, a new, preferable configuration will suddenly become admissible. This will increase L_{opt} to a new value, namely to the value of L_{tot} for the new configuration. If $L_{\max}$ continues to increase, then L_{opt} will again remain constant for a while, and then suddenly "jump" to a new value. Naturally, the exact same arguments apply to the case of *decreasing* $L_{\max}$; in this case, L_{opt} will decrease in discrete jumps. It would appear to be worthwhile to compute, for all possible values of $L_{\max}$ near the given one, the quantity L_{opt} as a function of $L_{\max}$.

In the attached Table 5, we have computed all of the discrete values of L_{opt} in a roughly 20 cm neighbourhood of 2040 cm. The table should be read as follows. Given a value of $L_{\max}$, the largest value of $L_{\max}$ on the table which is less than or equal to the given value should be consulted. Its corresponding quantity L_{tot} is the required value of L_{opt} for the given $L_{\max}$.

The values were actually found by working backwards. For each combination of crates of types 5, 6, and 7 whose total length was near to 302.7, a computer program (using modified brute force) considered each configuration that included this combination and all

Table 5
Variance of L_{opt} with L_{max}

#	Configurations							Car 2							L_1	L_2	L_{opt}	W_1	W_2	$W_{(s)}$	W_{tot}	L_{max}	
	Car 1																						
	C_1	C_2	C_3	C_4	C_5	C_6	C_7																
1	1	6	2	3	1	4	1	7	1	7	3	0	0	0	10200	10380	180	20580	36500	25500	11000	62000	10380
2	2	0	4	5	2	3	1	6	7	5	1	0	0	0	10200	10347	147	20547	25500	38500	13000	64000	10347
3	1	1	4	5	3	2	1	7	6	5	1	0	0	0	10200	10314	114	20514	28500	37500	9000	66000	10314
4	0	2	4	5	4	1	1	8	5	5	1	0	0	0	10200	10281	81	20481	31500	36500	5000	68000	10281
5	6	1	6	0	0	1	4	2	6	3	6	0	0	0	10200	10253	53	20453	27000	28000	1000	55000	10253
6	1	0	6	5	5	0	0	7	7	3	1	0	0	1	10200	10248	48	20448	30500	39500	9000	70000	10248
7	2	3	4	3	2	4	0	6	4	5	3	0	0	0	10200	10227	27	20427	34500	30500	4000	65000	10227
8	5	2	6	0	1	0	4	3	5	3	6	0	0	0	10200	10220	20	20420	30000	27000	3000	57000	10220
9	8	0	0	6	3	1	0	0	7	9	0	0	2	0	10197	10197	0	20394	33000	34000	1000	67000	10197
10	3	0	7	2	4	2	0	5	7	2	4	0	0	0	10180	10181	1	20361	34000	35000	1000	69000	10181
11	1	6	0	5	0	2	3	7	1	9	1	0	0	0	10167	10166	1	20333	29500	26500	3000	56000	10167
12	7	0	0	6	5	0	0	1	7	9	0	0	1	0	10164	1016	0	20328	37000	34000	3000	71000	10164
13	7	2	1	3	1	1	3	1	5	8	3	0	0	0	10149	10151	2	20300	31500	26500	5000	58000	10151
14	2	1	7	2	6	0	0	6	6	2	4	0	0	0	10147	10148	1	20295	39000	34000	5000	73000	10148
15	0	7	0	5	2	0	3	8	0	9	1	0	0	0	10134	10133	1	20267	34500	25500	9000	60000	10134
16	2	4	1	5	0	3	2	6	3	8	1	0	0	0	10107	10106	1	20213	27500	29500	2000	57000	10107

crates of types 1, 2, 3, and 4, without regard to length constraints. It then selected from among these the configuration that best balanced the two cars in the sense of minimizing $L_{|B|} = |L_{\text{car } 1} - L_{\text{car } 2}|$. For each such "minimizing" configuration, the value $\max(L_{\text{car } 1}, L_{\text{car } 2})$ represents the smallest value of L_{max} which allows the configuration to "fit," i.e., to satisfy the length restrictions R1 and R2. The results of the computer program can be sorted with respect to L_{max} and thus be converted into a table giving the largest obtainable value of L_{tot} for a given restriction value L_{max} .

We also attempted to analyze the stability of our solution with respect to changes in the crate data, but we found this to be more complicated than considering changes in the constraint values. Once again, the mass restriction is weak enough that small changes in the crate masses are unimportant. However, the effects of changes in the crate thicknesses can be dramatic. In a given optimal configuration, if the value of some l_i increases by more than $3 \text{ mm}/\max(n_i, n'_i)$ (which could be as little as 0.33 mm), the optimal configuration will be modified to the point of violating one of the restrictions R1 or R2; a new optimal configuration will then have to be sought. Similarly, if some l_i decreases by more than a very small amount (depending on i), then a given optimal configuration may have its L_{tot} decreased to the point where another admissible configuration is superior. If more than one of the l_i are changed, even smaller errors would make the solution unusable. If we assume that every thickness l_i increases by the same amount, then they each need increase by only 0.17 mm in order to cause our optimal configuration to violate a length restriction. This could happen, for example, if the temperature changed significantly. These considerations make it clear that our solution is very dependent on the values l_i . This must be admitted as a weakness of our solution which we have not rectified. That the weakness is significant is clear from the fact that the thicknesses of the crates are specified only to the nearest millimeter, and it must be assumed that the thicknesses have errors of up to 0.5 mm. In our recommended solutions 9, 14, 47 and 52, a change of 0.5 mm or less in any of l_1 , l_2 , l_3 , or l_4 could violate length restrictions, so our solution could not be used. In this case Table 1 could be used to find some alternative configurations worthy of consideration.

7. A Larger Problem

The solution of the given problem was greatly simplified by the strong constraint R5 (see Theorem 1). It is natural to consider what happens if R5 is removed. We noted above the difficulties with some

standard approaches. However, a number of reductions are possible, allowing us to use a modified brute-force approach. These reductions may be useful in other similar problems.

We are already aware that there exist good configurations for which L_{tot} is close to 2040 cm, since the optimal solutions for the old problem are admissible here too. Thus, $2039.4 \leq L_{\text{opt}}$ (for our new problem) ≤ 2040.0 . Assuming the mass constraints are still fairly weak (this assumption can be verified once a solution is found), we can simplify the new problem by considering only lengths.

Since L_{opt} is clearly an integer number of millimeters, with $2039.4 \leq L_{\text{opt}} \leq 2 \times L_{\text{max}} = 2040.0$, it must be one of the 7 values 2039.4, 2039.5, ..., 2040.0. We shall examine each of these 7 values in turn, beginning with the largest, 2040.0.

In our new problem, crates of type 1 and type 5 are the same (since they have the same length and we ignore mass), as are crates of type 2 and type 6. It suffices, then, to consider our problem as having 14 crates of type 1, 11 crates of type 2, and none of types 5 and 6. Other less obvious equivalences were searched for by a computer program, essentially by brute force. The simplest new equivalence found was: 2 crates of type 1 and 1 of type 7 are the same length as 4 crates of type 2. Unfortunately, for the small number of crates available in our problem, these equivalences are too unwieldy to be useful. Were there larger numbers of crates available, we could ignore all configurations with 4 or more crates of type 2, unless all crates of type 7 or all but one crate of type 4 were used. These equivalences could then yield substantial reductions.

Continuing with our new problem, we wish to satisfy the equation

$$487n_1 + 520n_2 + 613n_3 + 720n_4 + 640n_7 = 10200. \quad (1)$$

Searching for a specific optimum allows for number theoretic approaches. Consider (1) mod 10:

$$7n_1 + 3n_3 \equiv 0 \pmod{10}.$$

Multiply by 7:

$$-n_1 + n_3 \equiv 0 \pmod{10},$$

so

$$n_1 \equiv n_3 \pmod{10}.$$

Similarly,

$$n'_1 \equiv n'_3 \pmod{10}.$$

Now, we apply the symmetry constraint $n_1 \leq n'_1$. Since $n_1 + n'_1 \leq 14$, $n_1 \leq 7$. We also know that $n_3 \leq 9$, so (2) implies that $n_1 = n_3$.

Since $0 \leq n'_1 \leq 14$ and $0 \leq n'_3 \leq 9$, (3) reduces the possibilities for n'_1 to at most two for each n'_3 . For simplicity, assume that $n'_1 = n'_3$; if no solution is found, we can check the other possibility, $n'_1 = n'_3 + 10$. We must now satisfy:

$$(487 + 613)n_1 + 520n_2 + 720n_4 + 640n_7 = 10200,$$

which is the same as

$$55n_1 + 26n_2 + 36n_4 + 32n_7 = 510.$$

Considering this mod 2 shows n_1 is even, so let $n_1 = 2m_1$. After dividing our equation by 2, we have:

$$55m_1 + 13n_2 + 18n_4 + 16n_7 = 255,$$

with a similar equation for the primed variables. The availability constraints are now:

$$n_2 + n'_2 \leq 11,$$

$$n_4 = n'_4 > 6,$$

$$n_7 + n'_7 \leq 8, \text{ and}$$

$$m_1 + m'_1 = (n_1 + n'_1)/2 = (n_3 + n'_3)/2 \leq 4.$$

Notice that n_2 and n'_2 are determined by the equations we are trying to satisfy, so we needn't search through *all* the values given by the availability constraint. (This reduction was in fact used in the computer program when we considered the old problem.)

We can now make a brute force search of the configuration space given by the availability constraints on m_4 , m'_4 , n_7 , n'_7 , m_1 , and m'_1 . A number of solutions were found, and after checking the mass constraints, the following was found to be a solution of our new problem:

$$\begin{aligned} n_1 = 0, \quad n_2 = 6, \quad n_3 = 2, \quad n_4 = 5, \quad n_5 = 2, \quad n_6 = 0, \quad n_7 = 2, \\ n'_1 = 4, \quad n'_2 = 0, \quad n'_3 = 4, \quad n'_4 = 0, \quad n'_5 = 0, \quad n'_6 = 5, \quad n'_7 = 5, \end{aligned}$$

which fits the cars perfectly, with $L_{\text{tot}} = 2040.0$, and masses of 32500 kg and 27000 kg, respectively.

We have thus found an exact solution to a problem which at first appeared too large for a computational search. This example illustrates some useful techniques for solving problems of this nature.

References

- Hillier, Frederick S., and Gerald J. Lieberman. 1967. *Introduction to Operations Research*. San Francisco, CA: Holden-Day.
- Taha, Hamdy A. 1987. *Operations Research, An Introduction*. 4th ed. New York: Macmillan.